EEEE4138 Aerial Robotics

Flight Control II

Cascaded position and attitude control

State-space models and feedback design

From a position request to rotor thrust

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Recall the vertical controller

Downwards is positive: $\ddot{z} = g - T/m$.

Request acceleration u_z; set T_cmd = m(g − u_z).

With ideal thrust tracking, $\ddot{z} = u_z$.

For a fixed target: $u_z = k_p(z^d - z) - k_d\dot{z}$.

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Why use two connected control loops?

To accelerate horizontally, a conventional quadrotor tilts.

Outer loop: position/velocity → desired acceleration and tilt.

Inner loop: attitude/rates → turning moments.

Rotor thrusts deliver the requested force and moments.

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How a position request becomes motion



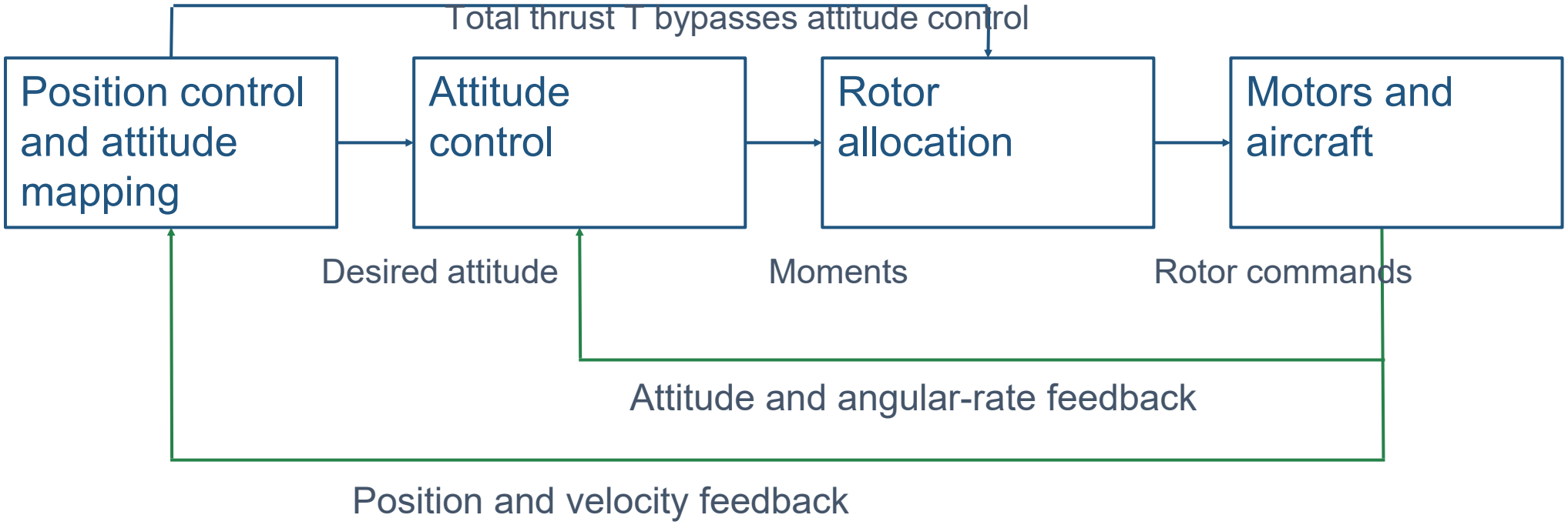
The aircraft cannot produce arbitrary horizontal thrust while staying level.

The inner loop must establish the tilt requested by the outer loop.

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The connected control architecture

Reference position and yaw enter the outer loop.



The attitude loop acts faster than the position loop.

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Start with the force equations

Use inertial x forwards and z downwards.

Positive pitch θ is nose-up; thrust tilts towards −x.

$$\ddot{x} = -(T/m)\sin\theta$$

$$\ddot{z} = g - (T/m)\cos\theta; \qquad J_Y\ddot{\theta} = M_Y$$

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Near hover : define the small changes

At level hover: $\theta_0 = 0$ and $T_0 = mg$.

Write $T = mg + \delta T$, where δT is a small change.

For small θ in radians: $\sin\theta \approx \theta$ and $\cos\theta \approx 1$.

Keep first-order changes; discard products of small changes.

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Near hover : substitute and simplify

$\ddot{x} \approx -[(mg + \delta T)/m]\theta = -g\theta - \delta T\theta/m.$

Discard the product $\delta T\theta$: $\ddot{x} \approx -g\theta$.

$\ddot{z} \approx g - (mg + \delta T)/m = -\delta T/m.$

Pitch mainly sets horizontal acceleration; thrust change sets vertical acceleration.

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Outer loop chooses an acceleration

$e_x = x^d - x; \quad \dot{e}_x = \dot{x}^d - \dot{x}$

$u_x = \ddot{x}^d + k_{px}e_x + k_{dx}\dot{e}_x$

Near hover: $\ddot{x} \approx -g\theta$.

Choose $\theta^d \approx -u_x/g$.

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What the outer loop symbols mean

x^d, x: target and actual position [m]

$\dot{x}^d$, $\dot{x}$: target and actual velocity [m/s]

u_x: acceleration request [m/s²]

θ^d: pitch request [rad]; k_{px} [s^{-2}], k_{dx} [s^{-1}]

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Inner loop makes the pitch request happen

Pitch error: $e\theta = \theta^d - \theta$.

For constant θ^d near hover: $\dot{e}\theta \approx -q$.

Angular-acceleration request: $\alpha_{cmd} = k_p\theta e\theta - k_d\theta q$.

Pitch moment request: $M_{\gamma,cmd} = J_\gamma \alpha_{cmd}$.

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Check whether the gains include inertia

Acceleration gains: $\alpha_{cmd} = k_p\theta e\theta + k_d\theta \dot{e}\theta$.

$k_p\theta = \omega_n^2$; $k_d\theta = 2\zeta\omega_n$.

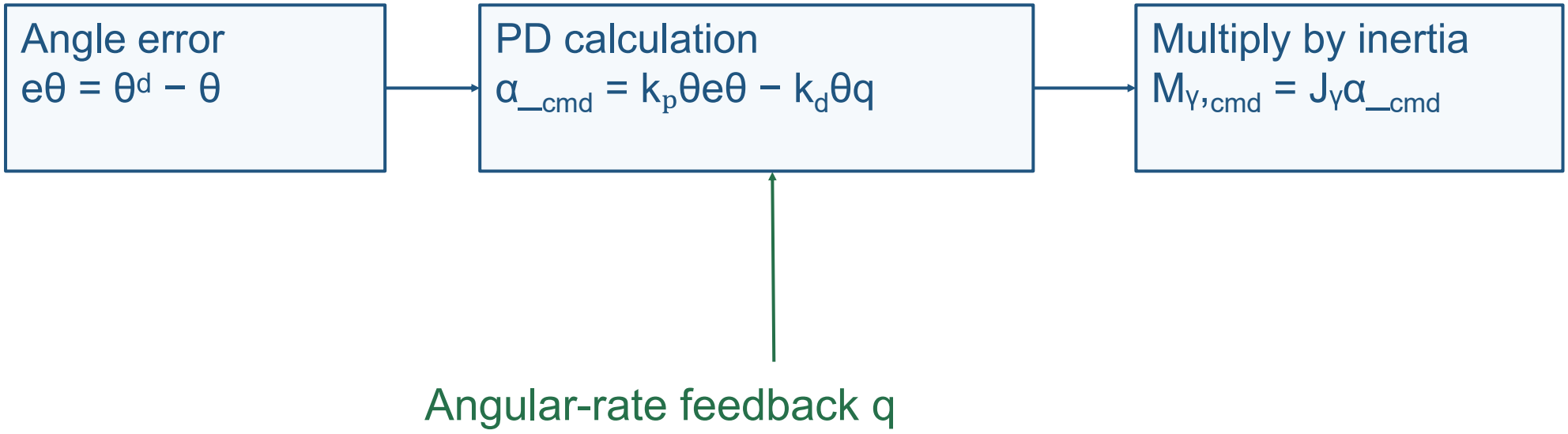
Torque gains: $M_{\gamma,cmd} = K_p\theta e\theta + K_d\theta \dot{e}\theta$.

$K_p\theta = J_\gamma k_p\theta$; $K_d\theta = J_\gamma k_d\theta$.

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Error and angular rate determine torque

For a constant pitch target near hover



Moment = rotational inertia × angular acceleration.

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Torque example — one calculation at a time

$\theta^d = 0.100$ rad, $\theta = 0$, $q = 0$.

$k_p\theta = 100\ s^{-2}$; constant reference gives $\dot{e}\theta = 0$.

$\alpha_{_cmd} = 100(0.100) = 10.0$ rad/s².

For $J_Y = 0.005$ kg m²: $M_{Y,cmd} = 0.0500$ N m.

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Why the attitude loop responds faster

Position control assumes the requested tilt appears promptly.

Use a faster inner attitude response as a design starting point.

Example natural frequencies: position 4, attitude 25 rad/s.

Then check motor lag, limits and the complete response.

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Worked example : attitude gains

$J_\gamma = 0.00500$ kg m²; $\omega_n = 25.0$ rad/s; $\zeta = 0.700$.

$k_p\theta = 625$ s⁻²; k_dθ = 35.0 s⁻¹.

$K_p\theta = 3.125$ N m/rad; $K_d\theta = 0.175$ N m s/rad.

With $\tau_m = 0.020$ s: 35.0 > $\tau_m(625) = 12.5$.

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Try it : position error to pitch moment

e_x = +0.0200 m; $\dot{e}_x$ = 0; k_{px} = 16.0 s^{-2}.

Use $\theta = q = 0$ and $\theta^{d} = \dot{\theta}^{d} = 0$ at this instant.

Use $K_p\theta$ = 3.125 N m/rad; $K_d\theta$ = 0.175 N m s/rad.

Calculate u_x, θ^{d} and $M_{y,cmd}$. Explain their signs.

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Solution : three linked calculations

$u_x = 16.0(0.0200) = +0.320$ m/s².

$\theta^d = -0.320/9.81 = -0.0326$ rad $= -1.87°$.

$e\theta = -0.0326$ rad; $\dot{e}\theta = 0$.

$M_{\gamma,cmd} = 3.125(-0.320/9.81) = -0.102$ N m.

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Maintain height while tilted

Tilt reduces the vertical component of total thrust.

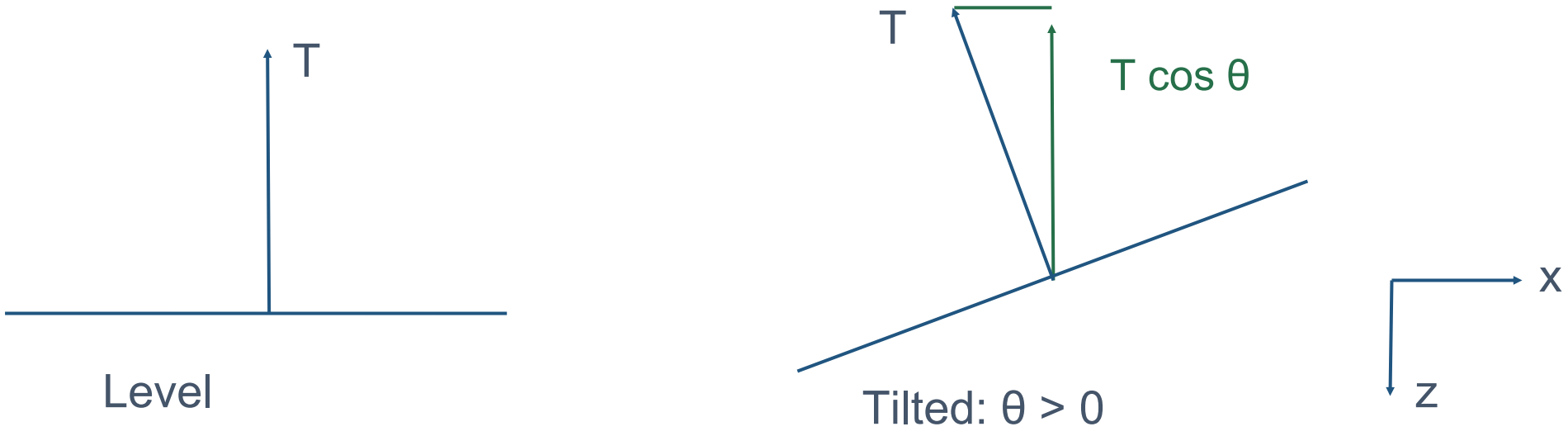
$\ddot{z} = g - (T/m)\cos\varphi\cos\theta$

To request u_z: $T_{cmd} = m(g - u_z)/(\cos\varphi\cos\theta)$.

Use only where tilt and rotor limits make this feasible.

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Why tilt needs extra thrust



Level: vertical thrust = T. Tilted: vertical thrust = T cos θ.

For zero vertical acceleration with roll zero: T = mg/cos θ.

At θ = 20°, required thrust is 1.064 mg: about 6.4% higher.

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Allocation converts loads to rotor commands

Plus layout: 1 front, 2 right, 3 rear, 4 left.

$$T = T_1 + T_2 + T_3 + T_4$$

$$M_x = l(T_4 - T_2); \quad M_y = l(T_1 - T_3)$$

$$M_z = k_{MT}(T_1 - T_2 + T_3 - T_4); \; \Omega_i = \sqrt{(T_i/kT)}.$$

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Feasibility matters as well as stability

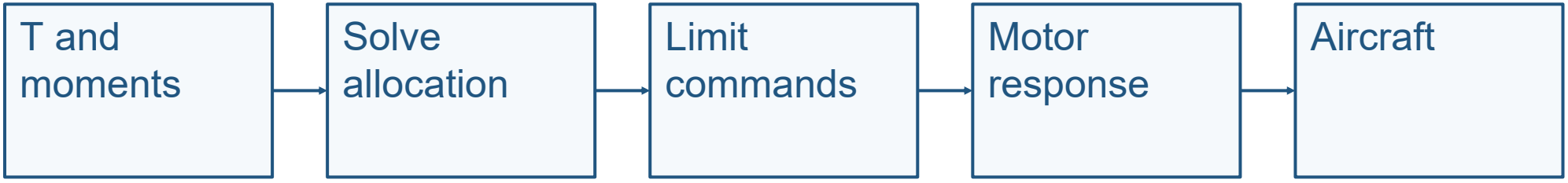
Each rotor must satisfy $0 \leq T_i \leq T_{i,max}$.

Motor lag delays even a feasible command.

Integral control must account for actuator limits.

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Allocation and actuators are part of the loop



A feasible rotor command still takes time to reach its target.

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A state records the present motion

To predict motion, position alone is insufficient.

We also need velocity.

Choose $\xi = [\delta z, \delta w]^T$ about stationary hover.

$\delta z = z - z_0$; $\delta w = w$; δ means deviation.

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Write two familiar equations together

$\delta\dot{z} = \delta w$; $\delta\dot{w} = v_c$, where $v_c = -\delta T/m$.

$\dot{\xi} = A\xi + Bv_c$; $y = C\xi + Dv_c$

$A = [0\ \ 1;\ 0\ \ 0]$; $B = [0;\ 1]$

For position output: $C = [1\ \ 0]$; $D = 0$.

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Multiply the matrices to recover the equations

First row: $\delta\dot{z} = 0(\delta z) + 1(\delta w) + 0(v_c)$.

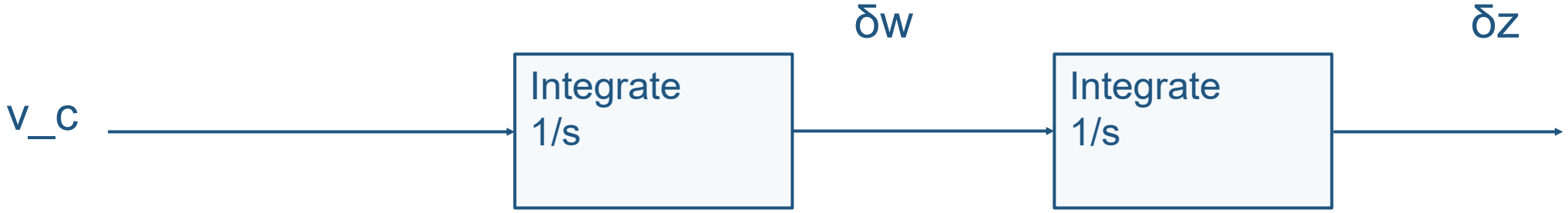
Second row: $\delta\dot{w} = 0(\delta z) + 0(\delta w) + 1(v_c)$.

Output: $y = 1(\delta z) + 0(\delta w)$.

Matrix form packages the equations; the physics is unchanged.

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The states are the integrator outputs



Input v_c [m/s²] → velocity deviation δw [m/s] → position deviation δz [m].

$\xi = [\delta z, \delta w]^T$: the vector order is position first, even though integration starts with acceleration.

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Input units determine the B matrix

Acceleration input v_c: $\delta\dot{w} = v_c$; $B = [0; 1]$.

Thrust input δT: $\delta\dot{w} = -\delta T/m$; $B_T = [0; -1/m]$.

A is unchanged: the underlying aircraft is the same.

A gain designed for one input cannot be copied to the other.

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Read A B C and D in words

A: current states → state derivatives.

B: selected input → state derivatives.

C: states → measured output.

D: immediate input → output; here D = 0.

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Controllability means inputs can move the states

For two states: controllability matrix = [B AB].

B = [0;1]; AB = [1;0].

[B AB] = [0 1; 1 0]; rank = 2.

Acceleration changes velocity, then position through integration.

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Observability means outputs reveal the states

Position output: C = [1 0].

CA = [0 1].

Observability matrix = [C; CA] = [1 0; 0 1].

Rank = 2: the ideal states can be inferred from output history.

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Feedback uses position and velocity

Choose $v_c = -K\xi = -k_1\delta z - k_2\delta w$.

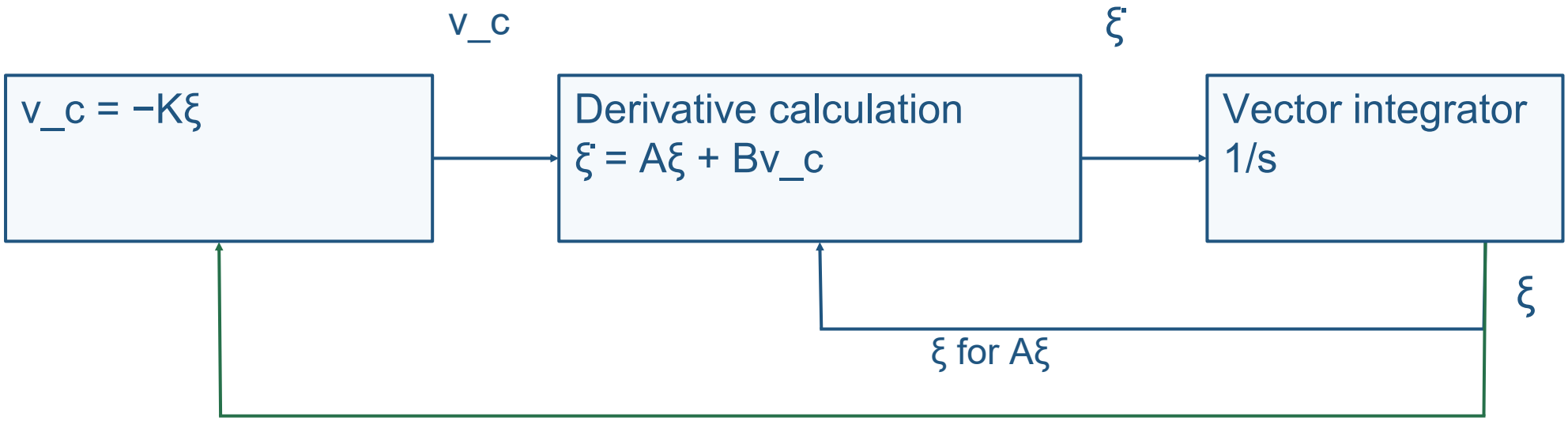
Then $\dot{\xi} = (A - BK)\xi$.

$A - BK = [0 \;\; 1; -k_1 \;\; -k_2]$.

Characteristic polynomial: $s^2 + k_2 s + k_1$.

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The state feedback loop includes integration



State feedback: position and velocity

Integration produces the states; the gain K uses their current values.

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Choose gains by choosing poles

Desired poles: -2 and $-3\ s^{-1}$.

$(s + 2)(s + 3) = s^2 + 5s + 6$.

Match $s^2 + k_2 s + k_1$: $K = [6 \quad 5]$.

For $\delta z = 0.100$ m, $\delta w = 0.200$ m/s: $v_c = -1.60\ m/s^2$.

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Turn the feedback command into thrust

$v_c = -1.60\ m/s^2$; $m = 1.50\ kg$.

$\delta T = -m v_c = +2.40\ N$.

$T_{cmd} = mg + \delta T = 14.715 + 2.40 = 17.115\ N$.

Upward acceleration requires thrust above weight.

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Track a target away from the hover point

r is desired position deviation from z_0.

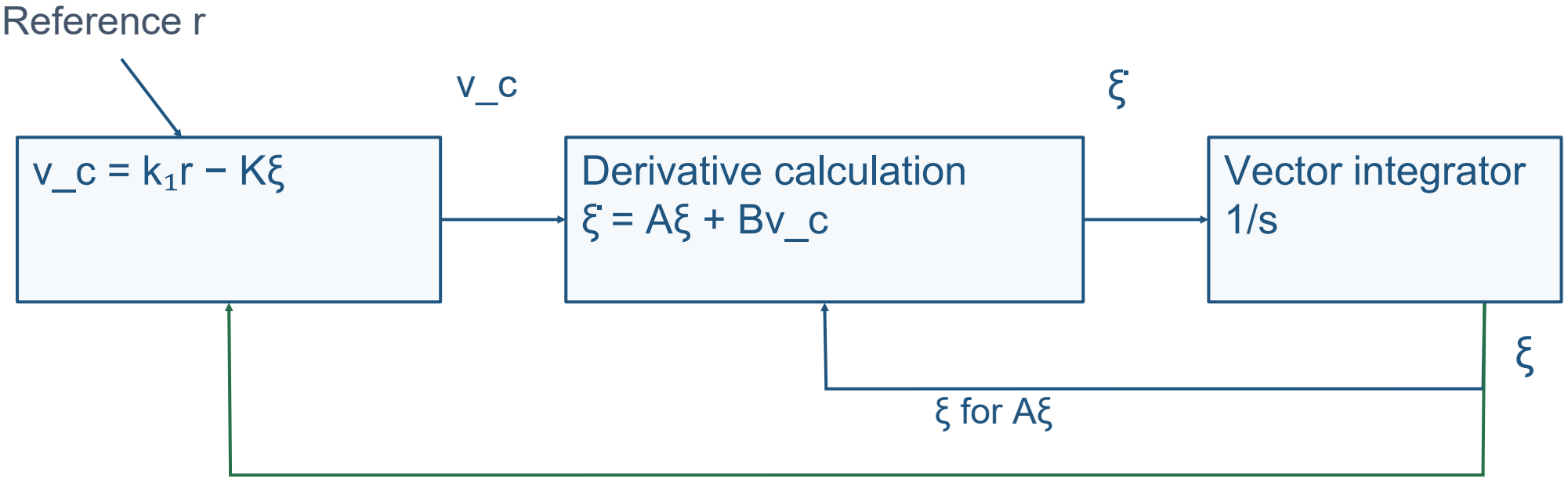
$v_c = k_1 r - k_1 \delta z - k_2 \delta w$.

Equivalently: $v_c = k_1(r - \delta z) - k_2 \delta w$.

At ideal steady state: $\delta z = r$ and $\delta w = 0$.

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Reference tracking and hover thrust



State feedback: position and velocity

Then convert acceleration to total thrust: T_cmd = mg − mv_c.

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Try it : state feedback design

Choose poles -3 and $-4\ s^{-1}$ for the two-state model.

Find K and $A - BK$.

At $\xi = [0,0]^T$ and $r = -0.200$ m, find v_c.

With $m = 1.00$ kg, find thrust and acceleration direction.

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Solution — state feedback design

$K = [12 \;\; 7]$; $A - BK = [0 \;\; 1; -12 \;\; -7]$.

$v_c = -12\delta z - 7\delta w + 12r$.

At the stated instant: $v_c = 12(-0.200) = -2.40$ m/s².

$T_{cmd} = 1.00(9.81 + 2.40) = 12.21$ N.

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Actuator lag introduces another state

Before: actual acceleration followed v_c ideally.

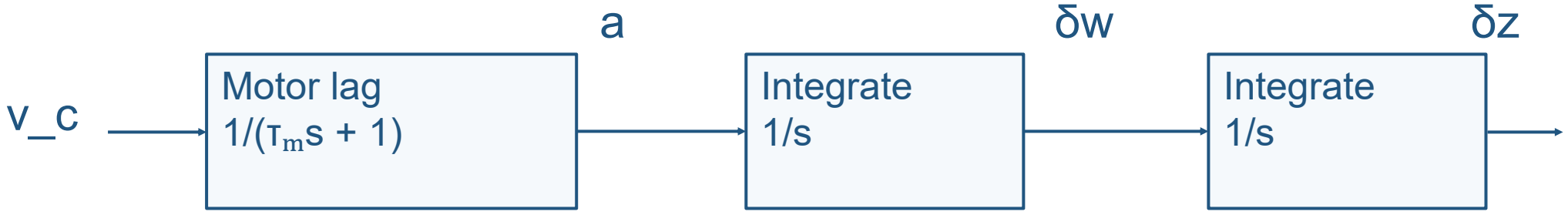
Now: $\tau_m \dot{a} + a = v_c$.

State vector: $\xi_a = [\delta z, \delta w, a]^T$.

Equations: $\delta \dot{z} = \delta w$; $\delta \dot{w} = a$; $\dot{a} = (v_c - a)/\tau_m$.

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Adding a motor state adds memory



v_c: commanded acceleration deviation; a: actual acceleration deviation.

$\tau_m \dot{a} + a = v_c \rightarrow$ actual acceleration catches up gradually.

States now are $\xi_a = [\delta z, \delta w, a]^T$.

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Extend the idea to the full aircraft

12 states: position, body velocity, attitude and body rates.

Linearise at a stated hover state and heading.

Inputs: δT, δM_x, δM_y and δM_z.

Moments influence horizontal motion through attitude.

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Four linked groups form the full model

Position	Velocity	Attitude	Angular rates

Position rates depend on velocity; horizontal acceleration depends on attitude.

12 states: 3 position + 3 velocity + 3 attitude + 3 body rates.

Four inputs: thrust deviation and three moment deviations.

At hover, moments affect horizontal position through attitude and acceleration.

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PD and state feedback can describe the same law

PD uses position error and its rate.

State feedback uses a vector containing position and velocity.

For this ideal channel, the two forms can be equivalent.

Cascaded control and state feedback can be used together.

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